

MANNVILLE AGCM, AB, Canada

WMO: 710470

Lat: 53.4556N

Lon: 111.2553W

Elev: 2152

StdP: 13.59

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-24.0	-18.9	-30.3	1.1	-23.8	-24.7	1.5	-18.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	82.0	62.4	78.9	61.5	76.2	60.7	66.9	77.2	64.6	74.6	62.7	72.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	93.1	72.6	60.6	85.5	69.9	58.6	79.5	67.7	32.7	76.9	30.8	74.5	29.3	72.8	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-30.1	87.5	2.7	2.2	-32.0	89.1	-33.6	90.4	-35.1	91.7	-37.1	93.3		
(5)			-30.6	70.9	2.9	2.3	-32.7	72.5	-34.4	73.8	-36.0	75.1	-38.2	76.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	11.7	10.2	22.3	37.4	52.2	58.9	62.3	60.9	52.7	39.0	22.8	9.3		
	DBStd	22.61	15.73	14.43	13.58	10.07	8.38	5.69	4.64	5.57	8.66	9.11	12.76	14.75		
	HDD50	6145	1188	1115	860	392	76	8	0	2	69	357	816	1262		
	HDD65	10373	1653	1535	1325	829	402	193	111	147	378	807	1266	1727		
	CDD50	1316	0	0	0	13	143	276	380	340	149	15	0	0		
	CDD65	70	0	0	0	0	4	11	27	21	7	0	0	0		
	CDH74	1087	0	0	0	5	132	147	304	312	184	3	0	0		
	CDH80	186	0	0	0	0	20	14	50	50	52	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	16.3	0.7	0.5	0.6	1.2	1.7	2.8	3.0	2.3	1.3	0.7	0.7	0.7		
(15) Precipitation	PrecMax	22.1	2.4	0.9	1.5	4.1	3.7	5.5	6.5	4.6	3.1	1.8	3.0	1.3		
	PrecMin	8.2	0.2	0.1	0.2	0.1	0.2	0.6	1.0	0.5	0.0	0.1	0.1	0.2		
	PrecStd	3.1	0.5	0.2	0.3	0.8	0.9	1.1	1.2	0.9	0.8	0.4	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	41.8	53.3	72.2	82.9	82.2	85.1	84.9	86.2	71.1	55.2	41.9		
		MCWB	36.8	36.0	42.8	50.1	56.2	61.5	67.0	64.7	60.7	53.8	43.8	34.9		
	2%	DB	40.1	37.8	46.4	65.5	77.9	78.2	80.7	80.7	79.3	62.0	47.0	38.2		
		MCWB	34.9	33.4	38.5	47.3	54.2	59.9	66.3	63.7	58.4	49.0	39.7	33.1		
	5%	DB	37.4	34.6	41.5	59.3	73.1	74.7	77.3	77.6	73.7	57.1	41.5	33.9		
		MCWB	33.3	31.2	36.1	44.2	52.7	58.8	64.5	63.1	56.7	45.9	36.4	30.2		
	10%	DB	34.0	31.5	38.2	53.4	68.7	71.3	74.2	74.0	68.0	52.5	37.8	29.4		
		MCWB	30.5	29.1	33.7	41.4	50.8	57.0	62.5	61.4	54.2	44.0	34.0	26.4		
	0.4%	WB	37.4	36.7	44.0	51.4	58.2	65.9	70.9	69.1	62.8	55.1	44.8	36.0		
		MCDB	42.3	41.4	52.0	71.6	77.4	77.5	80.7	79.0	80.0	69.4	52.8	40.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	33.8	38.9	47.6	55.8	62.8	68.0	66.3	59.9	51.0	40.2	33.1		
		MCDB	39.5	36.9	45.2	63.2	73.0	73.4	78.4	76.7	76.5	59.1	46.1	37.8		
	5%	WB	33.4	31.5	36.2	44.8	53.9	60.7	65.5	64.1	57.5	47.7	37.0	30.2		
		MCDB	37.2	34.5	41.0	57.9	69.4	71.0	74.8	74.5	72.0	55.4	41.0	33.8		
	10%	WB	30.6	28.9	34.1	41.9	52.2	58.8	63.5	62.2	55.1	44.7	34.3	26.5		
		MCDB	34.1	31.6	37.9	52.0	66.5	68.6	72.4	72.2	66.3	52.3	37.7	29.3		
	5% DB	MDBR	15.8	16.1	15.5	18.9	22.6	19.8	20.1	21.5	22.1	17.4	13.7	13.8		
		MCDBR	16.9	18.0	17.8	29.1	28.8	24.8	25.3	27.6	31.6	25.4	18.2	18.1		
		MCWBR	13.2	14.9	11.7	15.0	12.1	11.1	13.5	13.4	13.9	14.2	12.2	14.5		
		MCWBR	16.2	17.1	17.3	27.5	25.9	21.6	23.1	24.7	29.8	23.5	17.1	18.5		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.284	0.295	0.331	0.350	0.368	0.364	0.376	0.312	0.276	0.249	0.219		
		taud	2.316	2.244	2.347	2.361	2.356	2.334	2.370	2.312	2.475	2.548	2.446	2.359		
	Ebn at Noon		238	259	284	286	285	279	279	267	276	267	240	233		
		Edh at Noon	18	27	30	34	36	37	36	36	27	20	16	14		
(44) All-Sky Solar Radiation	RadAvg		311	647	1125	1487	1850	1887	1951	1592	1120	647	311	232		
	RadStd		33	47	89	129	129	137	106	117	122	91	34	23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page